Sri Lanka Institute of Advanced Technological Education

Higher National Diploma in Information Technology

Advanced Technological institute, Dehiwala

Individual project proposal

“Smart IT Asset Lifecycle & Security Compliance Management System”

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I. Introduction

In many organizations, IT assets such as computers, laptops, servers, and network devices are used every day. Managing these assets properly is very important for smooth business operations and security.

However, many companies still use manual methods or basic systems to manage their IT assets. This creates problems such as difficulty in tracking assets, data errors, and increased security risks.

Therefore, there is a need for a smart system that can manage IT assets throughout their lifecycle and also check whether they follow security rules.

II. Problem Statement

IT administrators and organizations face several problems when managing IT assets.

- It is difficult to track all assets properly
- It is unclear which employee is using which asset
- Maintenance records are not properly maintained
- Security rules (like antivirus and updates) are not always followed

Because of these problems:

- Assets can be lost or misused
- Security risks increase
- Time and resources are wasted

So, a centralized and automated system is needed to solve these problems.

III. Objectives

General Objective

To develop a smart web-based system to manage IT assets and ensure security compliance.

Specific Objectives

- To register and manage IT assets
- To assign assets to employees and track them
- To record maintenance and repair details
- To check security compliance (antivirus, updates, etc.)
- To generate reports and alerts

IV. Scope of the System

In Scope

- Asset registration and management
- Employee assignment tracking
- Maintenance and repair tracking
- Security compliance checking
- Alerts and notifications
- Report generation

Out of Scope

- Mobile application development
- Advanced IoT-based real-time monitoring
- Integration with external enterprise systems

V. Proposed Solution

The proposed system is a web-based application that allows organizations to manage all their IT assets in a centralized and efficient manner. It provides a user-friendly interface where administrators can easily monitor, control, and maintain assets throughout their lifecycle.

The system includes several key features:

- Asset Management Dashboard
  A central dashboard that displays all IT assets such as computers, laptops, servers, and network devices. It shows important details like asset status (active, in repair, retired), location, and usage, helping administrators quickly understand the overall asset condition.
- Employee Assignment System
  This feature allows assets to be assigned to employees. It keeps records of which employee is using which asset and supports transferring or returning assets. This improves accountability and reduces the risk of asset loss.
- Maintenance Tracking
  The system records maintenance and repair activities of each asset. It helps track service history, schedule repairs, and manage warranties, ensuring assets are properly maintained and have a longer lifespan.
- Security Compliance Monitoring
  The system checks whether each asset follows required security policies. This includes verifying antivirus installation, operating system updates, firewall status, and detecting unauthorized software. It helps maintain a secure IT environment.

- Alert System for Security Issues
  When a security problem is detected (for example, missing antivirus or outdated system), the system automatically generates alerts or notifications. This allows administrators to take quick action and prevent potential risks.
- Reports and Analytics
  The system generates detailed reports on asset usage, maintenance history, and security compliance. These reports help management make better decisions and improve overall IT asset management.

Overall, this system will help organizations improve efficiency, reduce manual work, enhance security, and manage IT assets more effectively, while also detecting and solving problems at an early stage.

VI. Methodology

The system will be developed using the **Waterfall model**.

Steps include:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment

VII. Technologies to be Used

Frontend

- React.js
- Tailwind CSS
- Chart.js
- Axios

Backend

- Node.js / ASP.NET Core

Database

- PostgreSQL

AI

- Python

VIII. Expected Outcomes

The system will:

- Improve efficiency in managing IT assets
- Reduce manual errors
- Improve security monitoring
- Provide real-time alerts
- Improve user experience

IX. Target Users

- IT Administrators
- Company Managers
- Employees

X. Timeline

- Week 1–2: Proposal
- Week 3–4: Requirement Analysis (SRS)
- Week 5–6: System Design
- Week 7–10: Development
- Week 11–12: Testing
- Week 13: Final Submission